

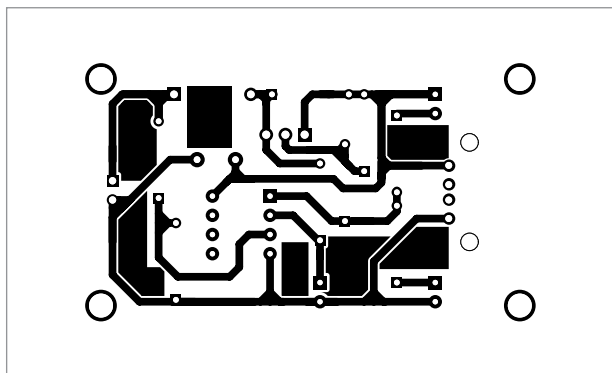


Fig. 3: Actual-size PCB layout for the battery charger

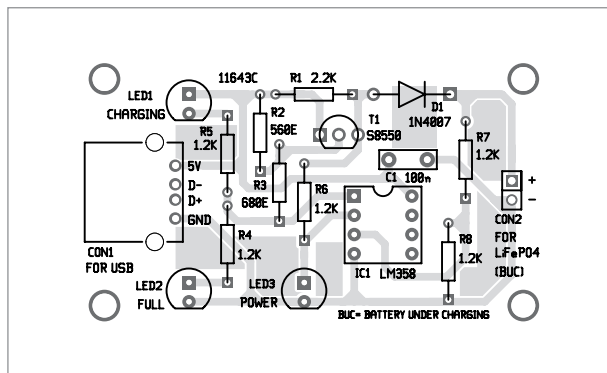


Fig. 4: Components layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - LM358 operational amplifier
- T1 - S8550 pnp transistor
- D1 - 1N4007 rectifier diode
- LED1-LED3 - 5mm LED (LED1 green in colour)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 2.2-kilo-ohm
- R2 - 560-ohm
- R3 - 680-ohm
- R4-R8 - 1.2-kilo-ohm

Capacitors:

- C1 - 100nF ceramic disk

Miscellaneous:

- CON1 - A-type USB
- CON2 - 2-pin connector
- BUC - LiFePO4 battery
- 18650 battery holder

cated by the glowing of LED2 and LED1's turning off. Later, when the battery voltage falls below the threshold level, the charger restarts a new charging cycle automatically, if the battery to be charged is in the holder connected to CON2.

Construction and testing

A small piece of perforated circuit board (or zero PCB) is enough for the construction of this circuit. An actual-size PCB layout for the little battery charger is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on a perforated board or PCB, connect a 5V DC supply across CON1. The author's prototype is shown in Fig. 5.

Use of a good quality single-cell 18650 battery holder is recommended for lodging the battery securely. You can make a nice enclosure for the LiFePO4 charger in case you wish to use it on long-term basis.

After construction, power the circuit (without the LiFePO4 battery) with a stable 5V DC supply provided

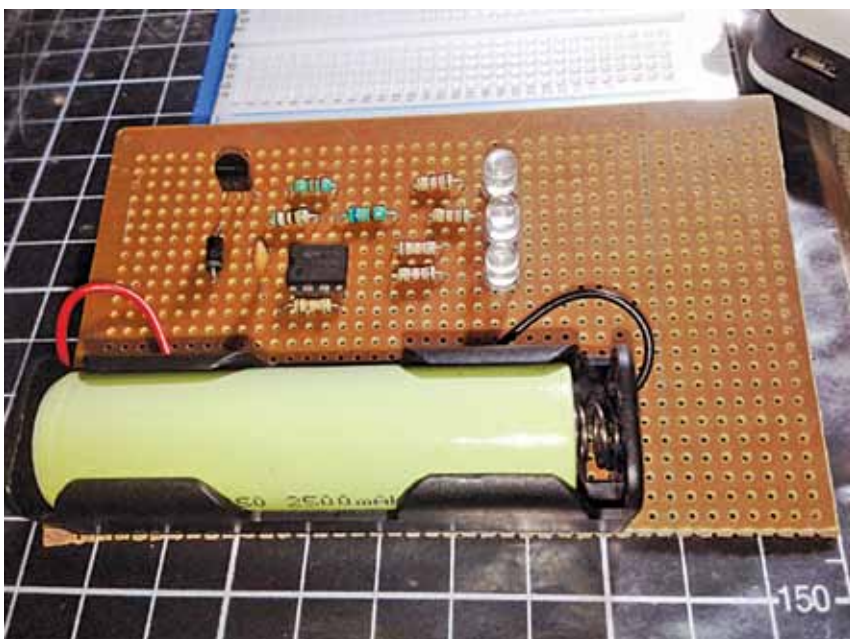


Fig. 5: Author's prototype

by a USB port, or a USB power adaptor, or a regulated lab power supply. If things are fine, you will see LED2 (full charge) and LED3 (power) glow. If you connect a voltmeter across the battery holder (CON2), it will display a DC voltage close to 3.6V. Now, when you insert the LiFePO4 battery into its holder, LED1 (charging) will glow but LED2 will go off.

Note. This circuit is simple but has some inherent limits as well, such as:

1. Since all components' values are critical, the circuit should be wired exactly as shown in the schematic.

2. The author's prototype was tested successfully with a 3.2V/2500mAh LiFePO4 battery and a 5V/700mA USB power source (having 3.6V maximum charging voltage and 250mA maximum charging current).

You may need to change the values of resistors R2, R7, and R8 slightly to fine-tune the charger circuitry.

3. Sometimes the charging and full charge indicators (LED1 and LED2) may not work as expected and a feeble (dim) glow may be seen. If that happens, just ignore it.

4. IC1 op-amp is used here as a comparator without hysteresis. Often this causes the charging indicator to briefly bounce back and forth (random flicker), especially when the battery voltage level crosses (and re-crosses) the threshold region. **EFY**



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